

Roni Singh

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PROFILE

Final-year IT student at MAKAUT (SGPA 8.32/10) with hands-on experience in ML security and predictive modelling. Built a malware classification system using explainable AI (SHAP/LIME) achieving high detection accuracy across CNN, Random Forest, and SVM architectures. Seeking an entry-level Software Engineer or ML Engineer role where I can deliver impact from day one.

TECHNICAL SKILLS

Languages Python, Java, C
ML / Data TensorFlow/Keras, Scikit-learn, NumPy, Pandas, Matplotlib, SHAP, LIME
Backend MySQL, Flask (REST APIs)
Tools Git, GitHub, Jupyter Notebook
Concepts Machine Learning, Deep Learning, XAI, LSTM, CNN, Feature Engineering
Soft Skills Teamwork, Problem Solving, Communication, Leadership, Time Management, Adaptability

PROJECTS

Malware Analysis & Classification – [Python](#) · [TensorFlow](#) · [Scikit-learn](#) · [SHAP](#) · [LIME](#) · [NetworkX](#) · [Angr](#)

- End-to-end detection pipeline on Windows PE files – extracting static (entropy, headers), dynamic (API traces), and graph-based (CFG) features for multi-modal classification.
- Trained CNN, Random Forest, and SVM models; achieved **F1 > 0.95** across 8 malware families on held-out test set.
- Integrated SHAP & LIME explainability, making model decisions auditable for security analysts.

Stock Price Prediction with LSTM – [Python](#) · [TensorFlow](#) · [Pandas](#) · [Matplotlib](#)

- Stacked LSTM on 5 yr OHLCV data; achieved **RMSE ~2.3%** on NIFTY50 stocks, outperforming ARIMA and linear regression baselines.
- Walk-forward validation eliminates look-ahead bias and ensures realistic out-of-sample metrics.

Digital Product Marketplace – [Next.js](#) · [React](#) · [TypeScript](#) · [Tailwind CSS](#) · [MongoDB](#)

- Built a full-stack e-commerce platform for selling digital products such as templates, greeting cards, UI kits, and design assets.
- Developed secure authentication, product listings, shopping cart, checkout, and instant download functionality after purchase.
- Implemented responsive UI with dark/light mode, category-based browsing, search, and product detail pages.
- Integrated payment workflow with Card and PayPal options and designed a modern user experience similar to leading digital marketplaces.

Payroll Management System – [Python](#) · [MySQL](#) · [Flask](#)

- Web-based payroll tool automating TDS computation, payslip generation, and batch processing for 50+ employees via a Flask front-end.
- Normalised relational schema (employees, departments, payroll_runs) with stored procedures for monthly runs.

EDUCATION

B.Tech – Information Technology | Maulana Abul Kalam Azad University of Technology, WB July 2022 – Present

- SGPA: **8.32 / 10.0** Coursework: DSA, DBMS, Machine Learning, Computer Networks

WBCHSE – Higher Secondary | Burdwan CMS High School June 2022

Score: 72%

WBBSE – Secondary | Ramharipur Ramakrishna Mission High School June 2020

Score: 86%

CERTIFICATIONS & COMMUNITY

- Qualified GATE 2026 in Computer Science and Information Technology (CS).
- Software Development Intern at CodTech Technology – gained hands-on experience in full-stack web development, UI design, and project collaboration using React, Next.js, and TypeScript.
- Core Member, HACKAUT Tech Club (2023 – Present).
- Organized hackathons, coding contests, workshops, and technical events as part of the college technology club.
- Coordinated registrations, promotions, scheduling, and participant support for events with 100+ participants.
- Contributed to building a stronger technical and developer community within the college.
- Member of HACKAUT college tech club; helped organize hackathons, coding contests, technical workshops, and campus events for 100+ participants.
- Actively participated in inter-college coding contests, programming competitions, and technical club events.